

Energy from Food

Bonus · Respiration (Cellular Energy)

Starring Aarav 🏃 & Glucose 🍬

A BAIO storybook by Narendra Sir · Biology Lecturer



Race Day

Aarav stood at the starting line of the school race. "Where will my muscles get their energy?" he wondered. Inside his cells, **glucose** from breakfast and **oxygen** from the air were about to begin a journey called **cellular respiration**: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O +$ energy. The energy is stored in **ATP**, the cell's energy currency.

💡 REMEMBER

Respiration is the breaking of C–C bonds of food by oxidation inside cells to release energy, trapped as ATP.

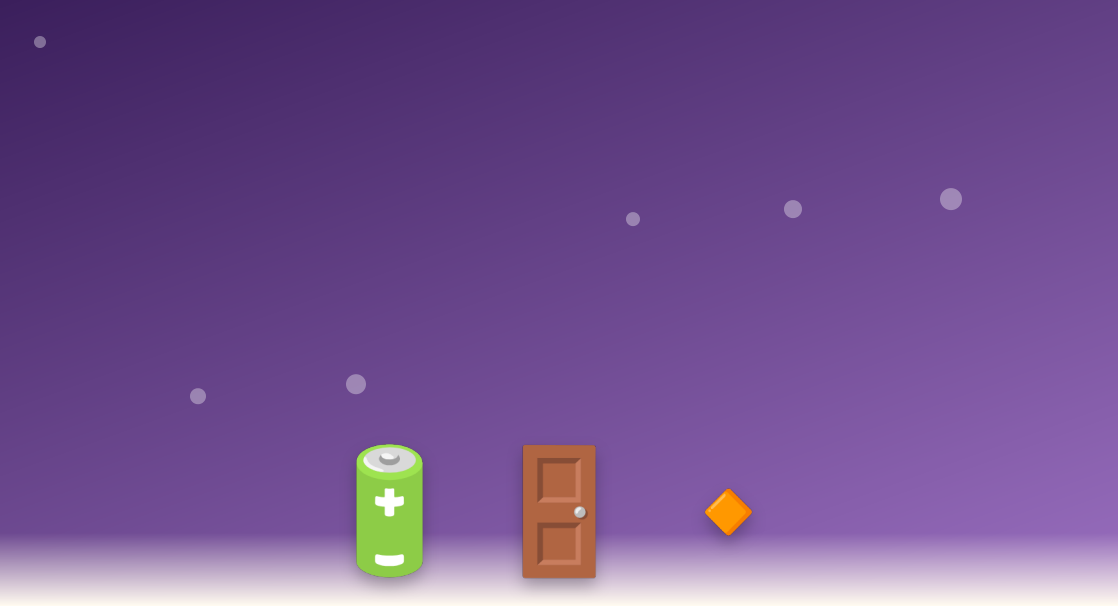


Splitting the Sugar

The journey began in the **cytoplasm** with **glycolysis**, also called the **EMP pathway** after Embden, Meyerhof and Parnas. One glucose (6C) was split into two molecules of **pyruvic acid** (3C). "No oxygen needed yet," said Glucose. The net gain was 2 ATP and 2 NADH.

💡 **REMEMBER**

Glycolysis occurs in all living organisms; the fate of pyruvic acid depends on whether oxygen is available.



Into the Powerhouse

With oxygen around, pyruvic acid entered the **mitochondria**. In the **matrix**, the enzyme **pyruvate dehydrogenase** converted it into **acetyl CoA**, releasing CO_2 and making NADH. This link step connects glycolysis to the next big wheel of energy.

💡 **REMEMBER**

Aerobic respiration takes place in the mitochondria and needs oxygen.



The Krebs Wheel

Acetyl CoA joined **oxaloacetic acid** to form **citric acid**, starting the **Krebs cycle**, discovered by **Hans Krebs**. As the wheel turned, CO_2 was released and the energy carriers **NADH** and **$FADH_2$** were filled up, along with one GTP. "The real energy cash is still waiting," said Glucose.

💡 **REMEMBER**

For each pyruvate, the Krebs cycle gives 3 NADH, 1 $FADH_2$ and 1 ATP (GTP), releasing 2 CO_2.



The Electron Waterfall

NADH and $FADH_2$ handed their electrons to the **electron transport system** (ETS) in the **inner mitochondrial membrane**. Electrons flowed down like a waterfall, and **oxygen** waited at the bottom as the **final electron acceptor**, forming water. The energy drove **ATP synthase** (F_0–F_1) to make ATP — **oxidative phosphorylation**.

💡 **REMEMBER**

One NADH gives about 3 ATP and one $FADH_2$ about 2 ATP; aerobic respiration yields a net of up to 38 ATP per glucose.



The Final Sprint

In the last 100 metres, Aarav's muscles needed energy faster than oxygen could arrive. So they switched to **anaerobic respiration**: pyruvic acid became **lactic acid** with the help of lactate dehydrogenase. Only 2 ATP per glucose — and the lactic acid made his legs ache with **fatigue**. Yeast does something similar, making **ethanol** and CO_2 — **alcoholic fermentation**.

💡 **REMEMBER**

Fermentation releases less than 7 percent of the energy in glucose.



The Respiratory Quotient

After the race, Aarav's teacher showed him the **respiratory quotient** (RQ) = volume of CO_2 released ÷ volume of O_2 used. "For carbohydrates RQ is 1. For fats it is less than 1 — about 0.7 for tripalmitin. For proteins it is about 0.9." Because respiration both breaks down and builds molecules, it is called an **amphibolic pathway**.

💡 **REMEMBER**

Fats enter respiration as glycerol and fatty acids; proteins enter as amino acids after deamination.



Across the Finish Line

Aarav crossed the finish line first! His cells had made enough ATP to power every stride. As he breathed deeply, oxygen rushed back to clear the lactic acid. "So that is why we breathe," he laughed. "Food gives the fuel, oxygen burns it, and mitochondria turn it into ATP!"

💡 **REMEMBER**

Aerobic respiration products: CO_2, water and energy (ATP).



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

glucose p.1 · oxygen p.1 · cellular respiration p.1 · ATP p.1

cytoplasm p.2 · glycolysis p.2 · EMP pathway p.2 · pyruvic acid p.2

mitochondria p.3 · matrix p.3 · pyruvate dehydrogenase p.3

acetyl CoA p.3 · oxaloacetic acid p.4 · citric acid p.4 · Krebs cycle p.4

Hans Krebs p.4 · NADH p.4 · $FADH_2$ p.4

electron transport system p.5 · inner mitochondrial membrane p.5

final electron acceptor p.5 · ATP synthase p.5

oxidative phosphorylation p.5 · anaerobic respiration p.6 · lactic acid p.6

fatigue p.6 · ethanol p.6 · alcoholic fermentation p.6

respiratory quotient p.7 · amphibolic pathway p.7



QUICK QUIZ

Do you remember?

1. Which organelle carries out aerobic respiration?

- Chloroplast
- Mitochondrion
- Nucleus
- Ribosome

2. What are the products of aerobic respiration?

- Glucose and oxygen
- CO_2, water and ATP
- Lactic acid only
- Ethanol and CO_2

3. Why do muscles produce lactic acid during intense exercise?

- Too much oxygen
- Shortage of oxygen leads to anaerobic respiration
- They make more glucose
- They stop respiring

4. Glycolysis takes place in the:

- Mitochondrial matrix
- Cytoplasm
- Nucleus
- Inner membrane

5. The final electron acceptor in aerobic respiration is:

- NAD^+
- Oxygen
- CO_2
- Pyruvate

Answers: 1-Mitochondrion · 2-CO_2, water and ATP · 3-Shortage of oxygen leads to anaerobic respiration · 4-Cytoplasm · 5-Oxygen